

Total No. of Questions : 8]

SEAT No. :

PC-1686

[Total No. of Pages :2

[6353]-2

T.E. (Civil)

Water Supply Engineering
(2019 Pattern) (Semester - I) (301002)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.
- 5) Use of scientific calculator is allowed.

- Q1)** a) Draw a neat sketch of a rapid sand gravity filter and show the various components. Explain the mechanics of rapid sand gravity filter? [6]
- b) A water treatment plant treats 200 m³/hr of water. Design the circular Clariflocculator. Following parameters are expected to be designed
i) Dimension of flocculator unit. ii) Power input by paddles to water.
iii) Size and number of paddles. iv) Opening below flocculator. [6]
- c) State the classification of coagulants? and explain any one. [6]

OR

- Q2)** a) Design six units of slow sand filter for the following data: [6]
- i) Population 60000
 - ii) Per capita demand 250 lit/cap/day
 - iii) Rate of filtration 180 lit/hr/sq. m.
 - iv) Length of each bed twice the breadth. Assume maximum demand of 1.8 the average daily demand, also assume that one unit is kept as a stand by.
- b) Compare the slow sand filter and rapid sand filter? [6]
- c) With neat sketch explain back washing of rapid sand gravity filter? [6]
- Q3)** a) Alum dose of 30 mg/l is used to. treat 60MLD of water workout quantity of alum required per month by the water treatment plant, Also calculate the amount of CO₂ released per day? [6]
- b) Define disinfection and list the different types of disinfection? [6]
- c) Explain Zeolite process with neat sketch. [6]

P.T.O.

OR

- Q4)** a) A filtered water discharge of 2 MLD has a chlorine demand of 5.8 mg/l. it is required to maintain a chlorine residual of 0.2 mg/l. determine the quantity of bleaching powder necessary for six months (chlorine available 25 %)? [6]
- b) Explain the lime-soda process? [6]
- c) State the various methods of removing excess fluorides from water? Explain any one in detail? [6]
- Q5)** a) Explain the water harvesting system? [4]
- b) Write the advantages of gravity and pumping system? [4]
- c) A town with population of 2 lakh is to be supplied with water supply daily at 200 lit per head. The pumping is done from 6 am to 6 pm. The variation in demand is as follows: [9]

Time of the Day	6 am - 9 am	9 am - 12 noon	12 noon - 3pm	3 pm - 6 pm	6 pm - 9 pm
Demand	40%	10%	10%	15%	25%

Determine the capacity of the service reservoir by mass curve method.

OR

- Q6)** a) Determine the balancing capacity of an ESR for a town having population 2 million and water supply rate of 280 lit/cap/1. the water is pumped continuously for 24 hr. breakup of demand is [8]

Time	Demand (lit-cap/1)
3 am - 9 am	80
9 am - 1 pm	50
1 pm - 7 pm	85
7 pm - 11 pm	30
11 pm - 3 am	35

- b) State the various methods of distribution system? Explain any one [4]
- c) What is the service reservoir? [5]
- Q7)** a) Draw with sketch WTP for swimming pool? [5]
- b) Explain the smart city mission and ATAL mission for rejuvenation and urban transformation mission? [7]
- c) State the various types of tools used for plumbing work? [5]

OR

- Q8)** a) What is Jal Jeevan mission? State its implication in rural India. [6]
- b) Draw any four types of sanitary pipe fittings? [5]
- c) Explain with sketches Sluice valve, presser valve, rdflux valve? [6]

